BASIC ELECTRONICS – TYPED NOTES (PART–2)

Clean typed version prepared from the supplied handwritten notes. Page order and topic sequence are retained; circuit drawings are described where the handwritten sketch is not reproduced.

1. Diode Applications / Half-Wave Rectifier

Diode applications: There are many applications of a p-n junction diode, such as a rectifier, diode clamp and clipper, etc.

The rectifier converts an AC voltage into a unidirectional output. The diode conducts during one half-cycle of the input and blocks current during the other half-cycle.

Half-Wave Rectifier

The half-wave rectifier is a circuit in which a diode is used to obtain a pulsating DC output from an AC input.

Working: During the positive half-cycle, the diode is forward biased and conducts. During the negative half-cycle, the diode is reverse biased and does not conduct.

For an ideal diode: $V_f = 0$ and $r_f = 0$.

Output waveform: positive half-cycles are passed and negative half-cycles are blocked.

Average Value of Half-Wave Rectified Voltage

$V_{avg} = (1 / 2\pi) \int_{0}^{\pi} V_m \sin\theta \, d\theta = V_m / \pi$.

Therefore, $V_{avg} = 0.318 V_m$.

Average Current

$I_{avg} = I_m / \pi$.

For a half-wave rectifier, the current flows only during one half-cycle.

Transformer-Coupled Rectifier

A transformer may be used to step the AC voltage up or down and to provide electrical isolation between the source and the rectifier circuit.

Advantages of transformer coupling: (1) It allows the source voltage to be stepped up or down as required. (2) It electrically isolates the supply from the rectifier circuit and reduces shock hazard.

2. Half-Wave Rectifier – Basic Results

For a half-wave rectifier: $I_{dc} = I_m / \pi$.

$V_{dc} = V_m / \pi$.

RMS value: $I_{rms} = I_m / 2$ and $V_{rms} = V_m / 2$.

Peak Inverse Voltage (PIV)

PIV is the maximum reverse voltage that a diode must withstand when it is reverse biased.

For an ideal half-wave rectifier, $PIV = V_m$.

Ripple Factor

The ripple factor indicates the amount of AC component present in the rectifier output compared with its DC component.

$r = \text{sqrt}[(I_{rms} / I_{dc})^2 - 1]$.

For a half-wave rectifier: $r = \text{sqrt}[(I_m/2)/(I_m/\pi)]^2 - 1 = \text{sqrt}(\pi^2/4 - 1) \approx 1.21$.

Hence, ripple factor ≈ 1.21, or about 121%.

Rectification Efficiency

$\eta = P_{dc} / P_{ac}$.

For an ideal half-wave rectifier, $\eta_{max} = 4 / \pi^2 \approx 0.406$, i.e. 40.6%.

Diode Power

$P_{dc} = I_{dc}^2 R_L$.

$P_{ac} = I_{rms}^2 R_L$ (with the appropriate circuit resistance included).

3. Diode Equation and Forward Current

Shockley Diode Equation

$$I = I_s [e^{(V_D / (\eta V_T))} - 1].$$

Here, I_s is the reverse saturation current, V_D is the applied diode voltage, η is the ideality factor, and V_T is the thermal voltage.

$$V_T = kT/q.$$

At room temperature (approximately 27 °C or 300 K), $V_T \approx 26$ mV = 0.026 V.

Example: Ge diode

A Ge p-n junction diode has a reverse saturation current of 5 μA at 300 K. Find the voltage required to obtain a forward current of 50 mA.

$I_s = 5 \times 10^{-6}$ A; $I = 50 \times 10^{-3}$ A; $\eta \approx 1$; $V_T \approx 0.026$ V.

$50 \times 10^{-3} = 5 \times 10^{-6} [e^{(V_D/0.026)} - 1]$.

$e^{(V_D/0.026)} \approx 10001$.

$V_D / 0.026 = \ln(10001) \approx 9.21$.

Therefore, $V_D \approx 0.239$ V ≈ 0.24 V.

Thus, approximately 0.24 V forward voltage is required for a 50 mA forward current in this example.

4. Practical Diode Model

For a practical silicon diode, the forward voltage drop is approximately 0.7 V. A practical diode can be represented by a threshold voltage plus a small bulk resistance.

$$V_D = V_F + I_D r_B.$$

Example

A silicon diode dissipates 2.5 W at a forward current of 1.5 A. Determine the forward voltage drop and bulk resistance.

$$V_D = P / I_D = 2.5 / 1.5 \approx 1.67 \text{ V}.$$

$$V_D = V_F + I_D r_B.$$

$$1.67 = 0.7 + 1.5 r_B.$$

$$r_B \approx (1.67 - 0.7)/1.5 \approx 0.65 \ \Omega.$$

Static and Dynamic Resistance

Static (DC) resistance is the ratio of diode voltage to diode current at a particular operating point.

$$R_{DC} = V_D / I_D.$$

Dynamic (AC or small-signal) resistance is the slope of the V-I characteristic at the operating point.

$$r_d = \Delta V_D / \Delta I_D.$$

For an exponential diode: $r_d = \eta V_T / I_D$.

5. Dynamic Resistance of a P-N Junction Diode

Dynamic resistance is also called AC resistance or small-signal resistance of a diode.

It is defined as the small change in diode voltage divided by the corresponding small change in diode current around a particular operating point.

$r_d = \Delta V_D / \Delta I_D$.

For an exponential diode: $r_d = \eta V_T / I_D$.

At about 27 °C, $V_T \approx 26$ mV, so $r_d \approx \eta(26\text{ mV})/I_D$.

The dynamic resistance is the slope of the V-I characteristic at the operating point.

Example

For $I_s = 5$ μA, $V_D = 0.6$ V, $T \approx 300$ K and $\eta = 2$, the forward current obtained from the diode equation is approximately 0.514 mA.

$R_{DC} = V_D / I_D \approx 0.6 / 0.514\text{ mA} \approx 1.17$ kΩ.

$r_d = \eta V_T / I_D \approx (2 \times 0.026) / 0.514\text{ mA} \approx 101$ Ω.

6. Half-Wave Rectifier – Numerical Example

For a half-wave rectifier supplied through a transformer, the secondary voltage and load resistance determine the maximum and average current.

Example from the notes: a 230 V, 50 Hz supply is applied to a transformer with turns ratio N1:N2 = 3:1. The load resistance is 10 kΩ and the transformer secondary resistance is taken as 10 Ω; the diode forward resistance is taken as 75 Ω in the numerical work.

- $V_{s(rms)} = 230/3 \approx 76.67$ V.
- $V_{sm} = \sqrt{2}\, V_{s(rms)} \approx 108.4$ V.
- $I_m = V_{sm} / (R_L + R_f + R_s)$.
- $I_{dc} = I_m / \pi$.
- $I_{rms} = I_m / 2$.
- $V_{dc} = V_{sm} / \pi$.

The notes obtain approximately $I_m = 10.75$ mA, $I_{dc} \approx 3.42$ mA and $V_{dc} \approx 34.52$ V for the stated values.

7. Rectification Efficiency with Practical Resistances

η = P_dc / P_ac.

For the half-wave case used in the notes: $\eta = [4/\pi^2] \times [R_L / (R_L + R_f + R_s)]$.

Because the transformer secondary resistance and diode forward resistance add extra loss, practical efficiency is lower than the ideal value.

The notes calculate an efficiency of approximately 40.1% for the numerical example.

Transformer Coupling – Advantages

(a) The source voltage can be stepped up or stepped down as required. (b) The AC source is electrically isolated from the rectifier, reducing shock hazard.

PIV Rating

For a half-wave rectifier, PIV of the diode is approximately V_m for the ideal circuit.

8. Full-Wave Rectifier

A half-wave rectifier allows unidirectional current during one 180° half-cycle. A full-wave rectifier allows current during both half-cycles, i.e. for the complete 360° input cycle.

There are two main types of full-wave rectifier: (i) centre-tapped full-wave rectifier (FWR), and (ii) bridge full-wave rectifier.

Centre-Tapped Full-Wave Rectifier

A centre-tapped full-wave rectifier uses two diodes connected to the secondary of a centre-tapped transformer.

The transformer provides the required secondary voltage. During the positive half-cycle, one diode is forward biased and the other is reverse biased. During the negative half-cycle, their roles interchange. Thus the load current remains in the same direction.

The circuit waveform contains positive pulses in both half-cycles.

9. Centre-Tapped FWR – Operation

Step 1: During the positive half-cycle of the input voltage, the upper end of the centre-tapped secondary becomes positive. Diode D1 is forward biased and D2 is reverse biased. Current flows through D1, the load resistance R_L and the centre-tap return path.

Step 2: During the negative half-cycle, the lower end becomes positive. D2 becomes forward biased and D1 becomes reverse biased. Current again flows through R_L in the same direction.

Therefore, the output is a full-wave pulsating DC waveform.

For a full-wave rectifier: V_dc = 2V_m / π.

I_dc = 2I_m / π.

V_rms = V_m / √2.

I_rms = I_m / √2.

Ripple Factor

r = sqrt[(I_rms/I_dc)^2 − 1] = sqrt[(π/(2√2))^2 − 1] ≈ 0.482.

Thus the full-wave ripple factor is approximately 0.482 (48.2%).

10. Full-Wave Rectifier – Form Factor and Efficiency

Form factor, FF = $V_{rms} / V_{dc} = (V_m/\sqrt{2})/(2V_m/\pi) = \pi/(2\sqrt{2}) \approx 1.11$.

Ripple factor = $\text{sqrt}(FF^2 - 1) \approx 0.48$.

For an ideal full-wave rectifier, $\eta_{max} = 8/\pi^2 \approx 0.812 = 81.2\%$.

PIV of Centre-Tapped FWR

For a centre-tapped full-wave rectifier, the non-conducting diode can be subjected to a reverse voltage equal to approximately $2V_m$ of the half-secondary voltage. Hence PIV is higher than in the bridge rectifier.

Practical PIV

For a practical silicon diode, the forward drop may be taken as about 0.7 V. The notes use the corresponding practical adjustment in PIV calculations.

11. Numerical – Half-Wave Rectifier

Example: A half-wave rectifier has a diode forward resistance of 0.3 Ω. The input voltage is 10 V (rms) and $R_L = 2$ Ω. Calculate I_{dc} and I_{rms} through the load.

$V_m = \sqrt{2} \times 10$ V.

$I_m = V_m/(R_f + R_L) = \sqrt{2} \times 10 / 2.3 \approx 6.15$ A.

$I_{dc} = I_m/\pi \approx 1.96$ A.

$I_{rms} = I_m/2 \approx 3.08$ A.

12. Numerical – Transformer-Fed Half-Wave Rectifier

Example: A 230 V AC supply is applied to a transformer of turns ratio 10:1. The secondary feeds a half-wave rectifier with $R_f = 20\ \Omega$ and $R_L = 1\ k\Omega$.

$V_{s(rms)} = 230/10 = 23$ V.

$V_{sm} = \sqrt{2} \times 23 \approx 32.53$ V.

$I_m = 32.53/(1000 + 20) \approx 31.89$ mA.

$I_{dc} = I_m/\pi \approx 10.15$ mA.

$I_{rms} = I_m/2 \approx 15.94$ mA.

$P_{dc} = I_{dc}^2 R_L \approx 103.02$ mW.

$P_{ac} = I_{rms}^2(R_L + R_f) \approx 259.16$ mW.

$\eta \approx P_{dc}/P_{ac} \times 100 \approx 39.75\%$.

$V_{dc} = I_{dc} R_L \approx 10.15$ V.

PIV of diode $\approx V_m \approx 32.53$ V for the ideal half-wave arrangement.

13. Numerical – Centre-Tapped Full-Wave Rectifier

Example: $R_L = 500\ \Omega$ and the resistance of each half of the secondary winding is $50\ \Omega$. Each diode has an idealized I-V characteristic corresponding to a forward resistance of $50\ \Omega$.

$I_m = V_m/(R_L + R_f + R_s)$.

$I_{dc} = 2I_m/\pi$.

$I_{rms} = I_m/\sqrt{2}$.

Ripple factor $= \text{sqrt}[(I_{rms}/I_{dc})^2 - 1] \approx 0.482$.

Efficiency $= [8/\pi^2] \times [R_L/(R_L + R_f + R_s)]$.

The notes obtain an efficiency of approximately 73.8% for the stated resistance values.

14. Centre-Tapped Transformer – Another Numerical

A centre-tapped transformer has a 220 V primary and a secondary rated at 12-0-12 V. It is used in a full-wave rectifier with $R_L = 100\ \Omega$. Determine the DC output voltage, DC load current and PIV rating.

V_m (one half of secondary) $= \sqrt{2} \times 12 \approx 16.97$ V.

$V_{dc} = 2V_m/\pi \approx 10.8$ V.

$I_{dc} = V_{dc}/R_L \approx 10.8/100 = 108$ mA.

PIV of each diode $\approx 2V_m = 2\sqrt{2} \times 12 \approx 33.94$ V.

15. Centre-Tapped FWR – Power, Regulation and TUF

Example: The RMS voltage from either end of the secondary to the centre tap is 30 V. The diode forward resistance is 2 Ω, half-secondary winding resistance is 8 Ω and R_L = 1 kΩ.

V_s = 30 V rms; $V_{sm} = \sqrt{2} \times 30 \approx 42.43$ V.

$I_m = V_{sm}/(R_L + R_s + R_f) \approx 42.43/(1000+8+2) \approx 42$ mA.

$I_{dc} = 2I_m/\pi \approx 26.74$ mA.

$P_{dc} = I_{dc}^2 R_L \approx 0.715$ W.

For the full-load voltage regulation calculation, the notes use the no-load and full-load output voltages and obtain approximately 0.97%.

Rectification efficiency: $\eta = P_{dc}/P_{ac} \approx 0.802$ (about 80.2%).

Transformer Utilization Factor (TUF) is defined as DC power delivered to the load divided by the AC rating of the transformer secondary.

16. Maximum Allowable Peak Input for FWR

Example: For a full-wave rectifier, the maximum allowable average DC current in each diode is 1 A. Assume identical diodes, negligible diode forward resistance, and $R_L = 100\ \Omega$.

For a centre-tapped full-wave rectifier, $I_{dc} = I_m/\pi$ for each diode.

If $I_{dc} = 1$ A, then $I_m = \pi$ A.

$V_m = I_m R_L = \pi \times 100 \approx 314.16$ V.

For the centre-tapped circuit, the PIV requirement is approximately $2V_m \approx 628.32$ V.

17. Bridge Rectifier

A bridge rectifier is a full-wave rectifier circuit using four diodes forming four arms of an electrical bridge.

The AC supply is connected to one diagonal of the bridge, usually through a transformer, and the rectified DC output is taken from the other diagonal.

Advantages of Bridge Rectifier

(1) It does not require a centre tap on the transformer secondary. (2) The PIV rating of each diode is lower than the centre-tapped arrangement. (3) Transformer utilization is generally better and the transformer can be smaller for the same load requirement.

Operation – Positive Half-Cycle

During the positive half-cycle, the appropriate pair of bridge diodes is forward biased and the other pair is reverse biased. Current flows through the load in one direction.

Operation – Negative Half-Cycle

During the negative half-cycle, the opposite pair of diodes conducts. The load current still flows in the same direction.

18. Bridge Rectifier – Equations

Two diodes conduct simultaneously in each half-cycle.

$V_{dc} = 2V_m/\pi$.

$I_{dc} = 2I_m/\pi$.

$V_{rms} = V_m/\sqrt{2}$.

$I_{rms} = I_m/\sqrt{2}$.

$I_m = V_m/(R_L + 2R_f + R_s)$.

$P_{dc} = I_{dc}^2 R_L = (2I_m/\pi)^2 R_L$.

$P_{ac} = I_{rms}^2(R_L + 2R_f + R_s) = (I_m/\sqrt{2})^2(R_L + 2R_f + R_s)$.

For an ideal bridge rectifier: $\eta_{max} = 8/\pi^2 \approx 81.2\%$.

Ripple Factor

$r = \text{sqrt}[(I_{rms}/I_{dc})^2 - 1] = \text{sqrt}[(\pi/(2\sqrt{2}))^2 - 1] \approx 0.482$.

PIV of Bridge Diode

In a bridge rectifier, the PIV rating required for each diode is approximately V_m for the ideal circuit, rather than $2V_m$ as in the centre-tapped full-wave rectifier.

19. Bridge Rectifier – Disadvantages and Practical PIV

Disadvantages noted in the notes: (1) The rectifier uses four diodes. (2) Two diodes conduct at a time, so the circuit has an additional forward voltage drop of approximately 2V_f. (3) As a result, the output voltage is slightly reduced.

Practical Silicon Diodes

Peak output voltage ≈ V_peak(sec) − 1.4 V (for two silicon diode drops).

PIV ≈ V_peak(output) + 0.7 V for the practical diode treatment used in the notes.

Advantages Compared with Centre-Tapped FWR

(1) The bridge transformer can be smaller than a centre-tapped transformer for comparable output power. (2) The PIV rating of each diode is lower, approximately V_m.

20. Numerical – Bridge Rectifier PIV

A transformer is specified to provide 12 V rms at the secondary from a standard 110 V AC primary. Find the peak output and the required PIV rating for each diode, using a two-diode conduction model.

$V_p(sec) = 12\sqrt{2} \approx 17$ V.

Peak bridge output $\approx 17 - 1.4 = 15.6$ V.

PIV of each diode $\approx 15.6 + 0.7 = 16.3$ V.

21. Numerical – Half-Wave Rectifier with Diode Resistance

Example: A half-wave rectifier has diode forward resistance $R_f = 0.3\ \Omega$, input voltage $V_{in(rms)} = 10$ V and $R_L = 2\ \Omega$.

$V_m = \sqrt{2} \times 10$ V.

$I_m = V_m/(R_f + R_L) = \sqrt{2}\times 10/2.3 \approx 6.15$ A.

$I_{dc} = I_m/\pi \approx 1.96$ A.

$I_{rms} = I_m/2 \approx 3.075$ A.

22. Numerical – Transformer-Fed Half-Wave Rectifier

Example: A 230 V AC supply is applied to a transformer with turns ratio 10:1. The secondary feeds a half-wave rectifier with $R_f = 20\ \Omega$ and $R_L = 1\ k\Omega$.

$V_{s(rms)} = 230/10 = 23$ V.

$V_m = \sqrt{2} \times 23 \approx 32.53$ V.

$I_m \approx 32.53/(1000+20) = 31.89$ mA.

$I_{dc} \approx 10.15$ mA; $I_{rms} \approx 15.94$ mA.

$P_{dc} \approx 103.02$ mW.

$P_{ac} \approx 259.16$ mW.

$\eta \approx 39.75\%$.

$V_{dc} \approx 10.15$ V.

PIV ≈ 32.53 V.

23. Full-Wave Rectifier – General Formula Sheet

Half-wave: $V_{dc} = V_m/\pi$; $I_{dc} = I_m/\pi$; $V_{rms} = V_m/2$; $I_{rms} = I_m/2$; $\eta_{max} = 40.6\%$; ripple factor ≈ 1.21; PIV $= V_m$.

Full-wave: $V_{dc} = 2V_m/\pi$; $I_{dc} = 2I_m/\pi$; $V_{rms} = V_m/\sqrt{2}$; $I_{rms} = I_m/\sqrt{2}$; $\eta_{max} = 81.2\%$; ripple factor ≈ 0.482.

Form factor of full-wave rectifier $= \pi/(2\sqrt{2}) \approx 1.11$.

Bridge rectifier: PIV per diode $\approx V_m$ (ideal); two diodes conduct in series, giving approximately $2V_f$ total forward drop in the practical silicon-diode case.

24. Key Definitions

Ripple Factor

Ripple factor indicates the magnitude of the AC component in the rectifier output relative to the DC component.

r = I_ac(rms)/I_dc = sqrt[(I_rms/I_dc)^2 − 1].

Form Factor

FF = RMS value / average (DC) value.

Rectification Efficiency

Rectification efficiency is the ratio of DC output power delivered to the load to the AC input power supplied to the rectifier.

η = P_dc/P_ac × 100%.

PIV

Peak Inverse Voltage is the maximum reverse voltage that a diode must withstand without breakdown.

Transformer Utilization Factor

TUF is the ratio of DC power delivered to the load to the AC rating of the transformer secondary.

25. Final Problem – Full-Wave vs Half-Wave Voltage

A full-wave voltage occurs in each half-cycle of the input voltage and therefore has a frequency twice the input frequency. A half-wave voltage occurs once in each input cycle and has the same frequency as the input frequency.

Peak Output of a Bridge Rectifier

For a transformer secondary rated at 12 V rms:

$V_{peak(sec)} = 12\sqrt{2} \approx 17$ V.

For two conducting silicon diodes, $V_{peak(output)} \approx 17 - 1.4 = 15.6$ V.

PIV rating of each diode $\approx V_{peak(output)} + 0.7 \approx 16.3$ V, according to the practical model used in the notes.

Typed Formula & Revision Sheet

The following formulas are the main results appearing repeatedly in the supplied notes:

- Half-wave: $V_{dc} = V_m/\pi$; $I_{dc} = I_m/\pi$; $V_{rms} = V_m/2$; $I_{rms} = I_m/2$.
- Full-wave: $V_{dc} = 2V_m/\pi$; $I_{dc} = 2I_m/\pi$; $V_{rms} = V_m/\sqrt{2}$; $I_{rms} = I_m/\sqrt{2}$.
- Half-wave efficiency (ideal) = $4/\pi^2 = 40.6\%$.
- Full-wave efficiency (ideal) = $8/\pi^2 = 81.2\%$.
- Half-wave ripple factor ≈ 1.21.
- Full-wave ripple factor ≈ 0.482.
- Full-wave form factor ≈ 1.11.
- Shockley equation: $I = I_s[e^{(V_D/(\eta V_T))} - 1]$.
- Thermal voltage: $V_T = kT/q \approx 26$ mV at room temperature.
- Dynamic resistance: $r_d = \eta V_T/I_D$.
- Practical silicon diode forward drop ≈ 0.7 V per conducting diode.

This typed version follows the supplied handwritten notes and keeps the same electronics topics: diode applications, half-wave rectification, diode equation, static/dynamic resistance, centre-tapped full-wave rectification, bridge rectification, PIV, ripple factor, efficiency, transformer coupling and numerical problems.